Analysis of the Temperature Properties of the Temperfect Mug

FYS2160 Oblig 1

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A : Temperfect Mug

1 Abstract

This report presents an analysis of the temperature of the Temperfect mug, which uses Phase Change Material (PCM) . The temperature curve of the Temperfect mug was compared with an ordinary mug. This analyzing found that the Temperfect mug keeps its temperature longer with more efficacy .

2 Introduction

In our fast-paced daily life, lovers of hot beverages, especially coffee lovers like myself, rely on consuming enough amount to keep our energy levels steady(which i call it "Dark Energy"). The temperature of these beverages is important: they shouldn't be so hot that they burn, nor so cold that they lose their comforting feeling. The Temperfect mug, as its name suggests, attempts to offer a "perfect" temperature for coffee. Unlike ordinary mugs that allow the drink to cool down rapidly or insulated thermoses that keep xcessive heat for longer time. This experiment aims to scientifically evaluate the claims made by the manufactures of the Temperfect mug by comparing its temperature capabilities with an ordinary mug. Simultaneously understanding how do they work.

3 Theory

TheTemperfect Technology: Temperfect Mugs add an extra layer of special "Temperfect" insulation to a standard double-wall travel mug. The Temperfect insulation quickly absorbs the excess heat from your freshly-brewed hot beverage. This means your drink will reach the perfect temperature fast, and you can start enjoying it right away. The excess heat is stored in the insulation layer and released over time to keep your drink at the optimum drinking temperature for hours (below). The Temperfect insulation is a phase-change material (PCM) . A wax-like substance which changes from solid to liquid phase as it absorbs heat, and from liquid back to solid as that heat is used to keep your drink at a perfect temperature.

There are two aspects of the physical process of phase change that make it perfect for this application:

1. The change occurs at a constant temperature. This property is what keeps your beverage at a constant temperature.

2. A large amount of energy must be transferred either to or from the "PCM" to change its phase. This is what enables it to store energy to keep your drink at just the right temperature for a long time.

4 Results

First, we open the data file "termokopper.txt" with Python and plot the temperature for the Temperfect mug and the ordinary one.
From the plot shown in Figure 1,we observe that the temperature of the ordinary mug steadily declines towards room temperature, which is the expected behavior. In contrast, the Temperfect mug displays a rapid decrease in temperature from 80°C to 65°C within a short time .Then it maintains the temperature close to 65°C for approximately 1800 seconds before it starts to keep the same pace as the ordinary mug to reach room temperature after approximately 6600 seconds.

Why does the Temperfect mug behave in this manner?

To understand this, we need to look into the phase transitions of (PCMs). In their default state, PCMs are solid. However, they melt and transition to a liquid state as they absorb heat. When we pour coffee at a high temperature

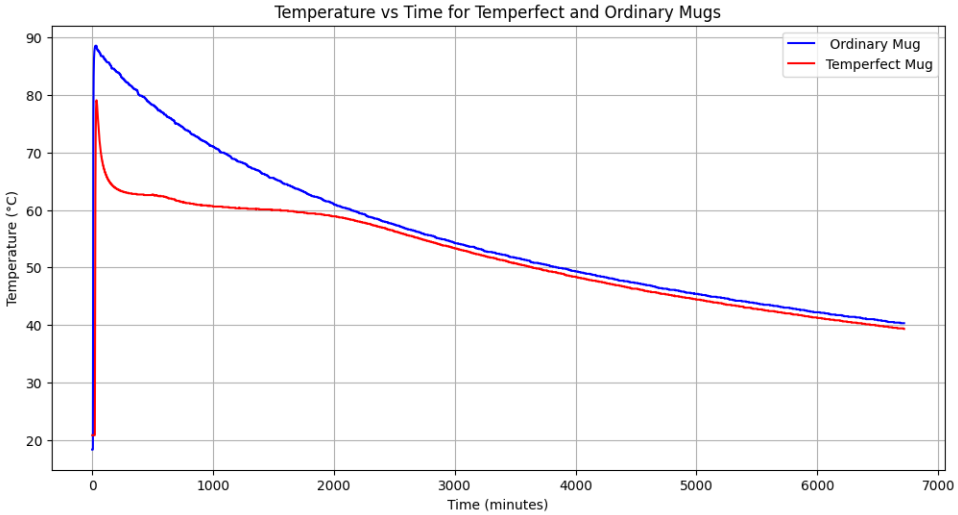


Figure 1: Temperature of the two mugs over time

into the mug, these materials absorb the excess heat, causing a phase shift to the liquid state. We know that liquids generally lose heat more slowly than solids (which is why areas close to the sea are warmer during winter compared to those further away).Thus, these PCMs release heat at a slower rate, maintaining the coffee close the ideal temperature.When the PCMs are no longer able to keep sufficient heat, they begin to shift back to their original solid state. This process is known as crystallization. Let us analyze the graph of the Temperfect mug .

We can see in Figure 2and 3 , the Temperfect mug has a rapid drop in temperature,namely from approximately 79°C to 62.8°C in 5 minutes. During this phase, the coffee transfers heat to the PCMs in the mug . It follows a phase of slower heat lose when PCMs are in the liquid state and it lasts for 35 minutes (from min 5 to min 40), until the temperature reaches56.8°C from 62.8°C.During this phase the heats flows back (from PCMs to the coffee). For the rest of time after (min 40) until (min 110), the Temperfect mug starts to behave as ordinary mug. Its temperature drops from 56.8°C to 38.8°C in in approximately 70 minutes. In this phase the PCMs are shifted back to their original solid state. We can identify :
-The melting Temperature is about62.8°C. -The melting period lasts about 35 minutes (ranging from 62.8°C to 56.8°C). - The crystallization begins at

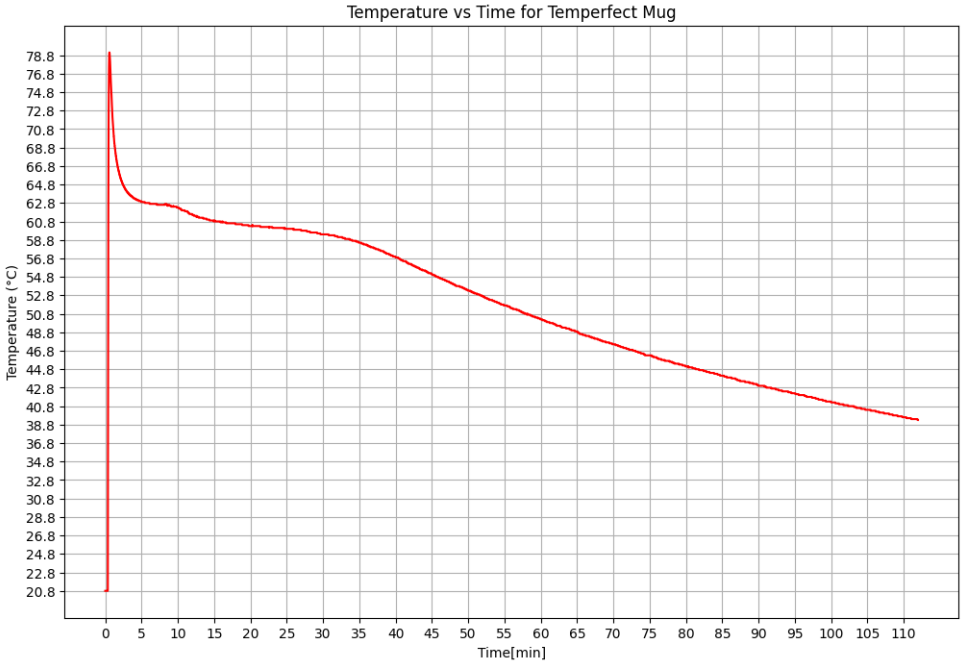


Figure 2: Temperature of Temperfect mug over time

56.8°C - The heat storing begins at 79°C until 62.8°C(The melting point). To find the heat stored during this phase,we write:

$$\Delta Q = c_{v_w} m_w \Delta T = 4.186 J/g 500 g (79 - 62.8°C) = 33906.6 \text{ J} = 33.9 \text{ kJ}.$$

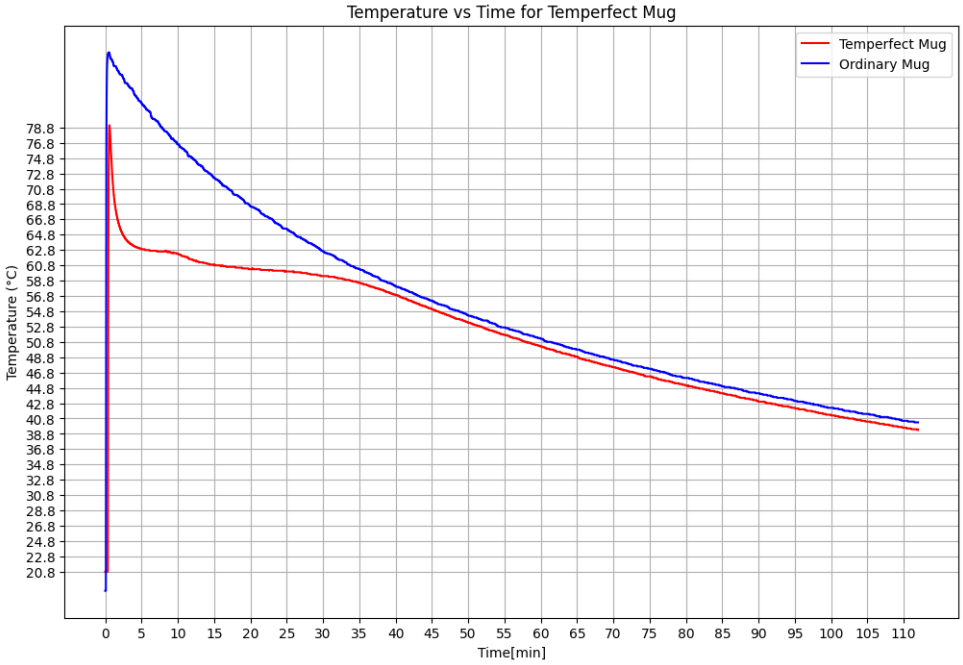


Figure 3: Temperature of ordinary and Temperfect mug over time

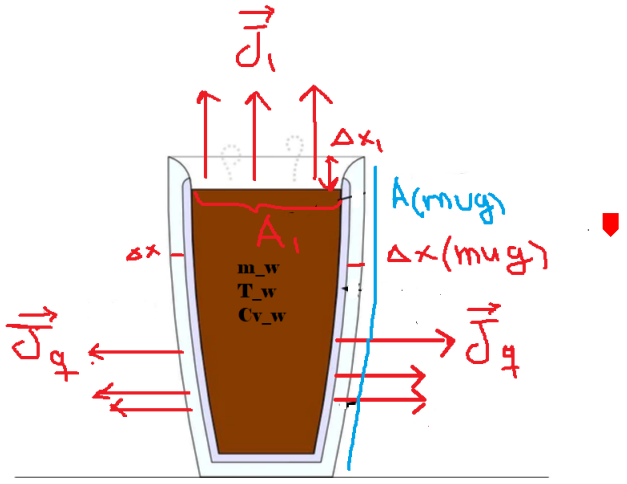


Figure 4: Sketch of the ordinary mug

In the ordinary mug, the heat flux is greater at the sides(from water to

the mug) and the top(from water to the air) . As the water at the top and sides cools, it becomes denser and sinks to the bottom, causing the warmer water to flow up as shown in figure 4. From the heat conduction equation :

$$\vec{J_q} = -\lambda \frac{\Delta T}{\Delta x} \tag{1}$$

The total flux in this case is the sum of the two fluxes : $\vec{J_q}$ and $\vec{J_1}$. We can rewrite the equation :

$$\frac{dQ}{dt} = A_{tot}\vec{J}_{tot} = A_{mug}J_q + A_1 J_1 \tag{2}$$

We substitute (1) in (2), we get :

$$\frac{dQ}{dt} = -A_{tot}\vec{J}_{tot} = -(A_{mug}J_q + A_1 J_1) = -\left(\frac{A_{mug}\lambda_{mug}\Delta T}{\Delta x} + \frac{A_1\lambda_1\Delta T}{\Delta x_1}\right) \tag{3}$$

We have the heat lose from water : $\frac{dQ}{dt} = m_w v_v w \frac{dT}{dt}$ and $T = \Delta T \implies \frac{dT}{dt} = \frac{d\Delta T}{dt}$ We can write equation (3):

$$m_w v_v w \frac{d\Delta T}{dt} = -\left(\frac{A_{mug}\lambda_{mug}\Delta T}{\Delta x} + \frac{A_1\lambda_1\Delta T}{\Delta x_1}\right) \implies \frac{d\Delta T}{dt} = -\frac{1}{m_w v_v w}\left(\frac{A_{mug}\lambda_{mug}}{\Delta x} + \frac{A_1\lambda_1}{\Delta x_1}\right)\Delta T \tag{4}$$

By taking dt to the right and bringing ΔT to the left , we get:

$$\frac{d\Delta T}{\Delta T} = -\frac{1}{m_w v_v w}\left(\frac{A_{mug}\lambda_{mug}}{\Delta x} + \frac{A_1\lambda_1}{\Delta x'}\right)dt \tag{5}$$

We define $\frac{1}{\tau} = \frac{1}{m_w v_v w}\left(\frac{A_{mug}\lambda_{mug}}{\Delta x} + \frac{A_1\lambda_1}{\Delta x'}\right)$.
The equation (5) becomes : $\frac{d\Delta T}{\Delta T} = -\frac{1}{\tau}dt$. By integrating both sides :

$$ln\Delta T = -\frac{t}{\tau} + C \implies \Delta T = e^{-\frac{t}{\tau}}e^C . t = 0 \implies \Delta T = \Delta T_0 e^{-\frac{t}{\tau}} \implies \frac{\Delta T}{\Delta T_0} = e^{-\frac{t}{\tau}} \tag{6}$$

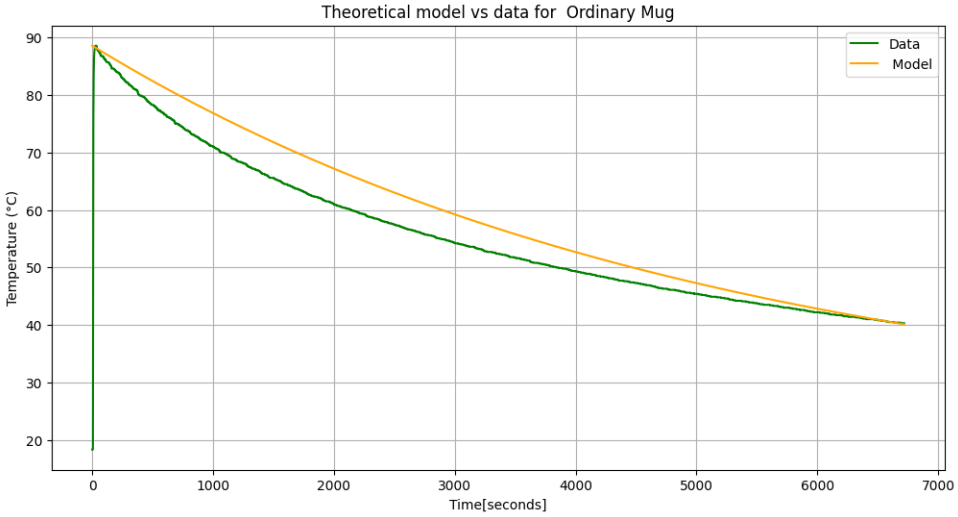


Figure 5: Plot of the model vs Data for ordinary mug

5 Methods

We used a python code that loads the txt file and analyze data as following :

Listing 1: Temperfect cup analyzing code

```
import matplotlib.pyplot as plt
import numpy as np


data = np.loadtxt("termokopper.txt", delimiter='\t')
time = data[:, 0]
temperature1 = data[:, 1]
temperature2 = data[:, 2]
time = time / 60

plt.figure(figsize=(12, 6))
plt.plot(time, temperature1, label="temperature1", color="blue"
    ↪ )
plt.plot(time, temperature2, label="temperature2", color="red")
plt.xlabel("Time[seconds]")
plt.ylabel("Temperature␣(C)")
```

```
plt.legend()
plt.grid(True)

plt.show()

"""
By plotting the data we could identify that temperature1 was for ordinary mug and
temperature2 is for Temperfect mug
"""

Ordinary_Mug = temperature1
Temperfect_Mug= temperature2


plt.figure(figsize=(12, 6))
plt.plot(time, Ordinary_Mug, label="Ordinary Mug", color="blue" )
plt.plot(time, Temperfect_Mug, label="Temperfect Mug", color=" red")
plt.xlabel("Time[seconds]")
plt.ylabel("Temperature (C)")
plt.title("Temperature vs Time for Temperfect and Ordinary Mug" )
plt.legend()
plt.grid(True)

plt.show()
"""
Then the Temperfect mug was plotted with more detailed way to analyze its properties .
"""
plt.figure(figsize=(12, 8))
plt.plot(time, temperature2, label="Temperfect Mug", color="red ")
plt.plot(time, temperature1, label="Ordinary Mug", color="blue" )

```

```
47
48 plt.xticks(np.arange(min(time), max(time), 5))
49 plt.yticks(np.arange(min(temperature2), max(temperature2), 2))
50
51 plt.xlabel("Time[min]")
52 plt.ylabel("Temperature␣(C)")
53 plt.title("Temperature␣vs␣Time␣for␣Temperfect␣Mug")
54 plt.grid(True)
55 plt.legend()
56 plt.show()
57
58 """
59 For the fiiting model :
60 """
61 tau = 6600
62 fit_model = (np.max(temperature1)-Ta)*(np.exp(-time/tau)) + Ta
63
64
65
66 plt.figure(figsize=(12, 6))
67 plt.plot(time, temperature1, label="Data", color="green")
68 plt.plot(time, fit_model, label="␣Model", color="orange")
69
70 plt.xlabel("Time[seconds]")
71 plt.ylabel("Temperature␣(C)")
72
73
74 plt.title("Theoretical␣model␣vs␣data␣for␣␣␣Ordinary␣Mug")
75 plt.legend()
76 plt.grid(True)
77
78 plt.show()
```

6 Conclusion

Based on the data ,we can see an evidence that the Temperfect mug has advantage in temperature regulation over ordinary mugs by keeping beverages at the optimal drinking temperature for longer time. This performance can be mainly related to the Phase Change Materials (PCMs) and their ability to absorb heat and release it back to the beverage and extending the relaxation time.

B : Algorithmic models

From the code "diffusion model start.py", we can identify the parameters related to thermal conductivity :

-Ctb: is the heat capacity ratio from top to bottom.In another words , it decides how much heat the lower block will absorb from the above one.A higher value means more absorbing .
-DT : the temperature difference between the top and bottom . The greater the difference, the more heat will transfer from top to bottom . To simulate the heat loss to the environment,we need to add heat lose factor $HL_{fac}$to the code . the heat lose factor will represent the heat lose from the top block to the surrounding room . We add the new variableDT_s which is the difference in temperature between the top and the surrounding room. if our random number is less than DT_s the top object will lose heat to the room which is more real than being the top cooler than the room. If the top is cooler than the room it will absorb heat and transfers it to the bottom.

Listing 2: adding heat loss to the environment

```
# we keep the first loop, then we add this:
if surrounding:
      r = 4 * np.random.rand(1, 1) - 2

      DT_s = Tt[i] - Tr
      if r < DT_s:
          Tt[i] = Tt[i] - HL_fac / N
```

The algorithmic model describes the transfer of heat between two blocks : top and bottom. By generating a random numbers which represent the random

flow of heat(as i understood it). if the number is less than the temperature difference between top and bottom, the heat will flow from top to bottom. This code use the principle of heat flow from the hotter to the cooler. Our experiment also examines the heat flow from a mug of coffee (the hotter) to the surrounding room(cooler).

II Appendix : Concept questions

1

a:

pressure in this case should increase due to the decreasing in volume. Mathematically :The expression $V^{\lambda}P$ = constant for Adiabatic compression . In order to keep $V^{\lambda}P$ = constant, pressure should increase as volume decreases.

b:

The temperature should increase because we did a positive work on the system . Mathematically : the internal energy U increases due the its proportionality with work. $\Delta U = U_2 - U_1 = \frac{f}{2}nR(\Delta T = T_2 - T_1)$. When $U_2 - U_1$ is positive, then $\Delta T = (T_2 - T_1) > 0$, $T_2 > T_1$. Thus T increases.

c:

As the volume decreases, the particles are closer , making the gas denser. This causes an increased number of collisions, that means a rising in both pressure and temperature. The kinetic energy of the molecules also increases as they move faster. As the kinetic energy rises, the internal energy of the gas also increases, and the work done on the gas is thus positive.

d:

No other factors will be changed.

2

a:

In this case, we have Isothermal compression. As we compress ,we do work on the system and the molecules will still experience more collisions but the energy remains unchanged and the work we do will compensate by the heat flow to the surrounding.Thus the temperature will not increase.

b:

In this case the pressure will increase as the volume decreases.

c:

The molecules will collide more frequently causing the pressure to increase. Since the temperature remains unchanged due to the heat flux to the surroundings, the speed of molecules will not changed either and no change in the kinetic energy.

d:

The density of the gas will increase due to the decreasing in volume .

3

a:

From the formula of energy of a gas: $U = \frac{f}{2}NkT$. The energy per molecule will be : $\frac{U}{N} = \frac{f}{2}kT$. Since the air and the water are in equilibrium , $T$is the same for both. the only factor that determines the energy per molecule is f(degrees of freedom). the air is diatomic gas with $f = 5$ and water has $f = 9$. The water has more energy per molecule than air.

b:

Water has more energy per molecule tan air this means water has more potential energy than air .

c:

From formula:$\frac{U}{N} = \frac{f}{2}kT$, we can see that the energy is proportional to f(degrees of freedom).Water molecules have more kinetic energy than air molecules.

d:

When two objects with two different temperature are in contact, heat tends to flow spontaneously from the object with high temperature to the object with lower temperature until both reach the thermal equilibrium.When water and air are in contact inside the the bottle and they are in thermal equilibrium, they will be at the same temperature.

e:

When we slowly heat both water and air, keeping them at the same temperature, the volume of both expands. The molecules in both move faster and gain more kinetic energy. This leads to more frequent collisions between molecules, resulting in an increase in pressure.

f:

When we put a glass of water filled with ice at room temperature, the ice will keep absorbing heat from the water and from the surrounding air and cooling the water around ice cubes. The ice will use this heat to melt while keeping it temperature at 0°C until all ice melts down and become water.the melted ice(water now) will continue to absorb heat from the water in the glass until all together reach the thermal equilibrium with surroundings at room temperature.

References

[1] Daniel V. Schroeder. *An Introduction to Thermal Physics*. Addison Wesley Longman, 2000.

[2] Anders Malthe-Sorenssen. *Elementary Thermal and Statistical Physics Using Python*. Springer, 2016.

[3] https://joeveo.com/pages/the-temperfect-mug